

TITLE:

Date:

C03117 UNDERSTAND (Perception + Delta)

Date: 28/09/2026

1) Perception update:

A perceptron is a linear function that classifies the input by computing the score using $z = w^T x + b$. When it misclassified, the perceptron updates its parameters using $w = w + \eta xy$, $b = b + \eta xy$ where η is learning rate.

2) Linear separability:

A dataset is linearly separable if a single linear line can separate the data into correct class.

3) Why XOR fails: The XOR problem is not linearly separable because its positive and negative classes cannot be classified using a single straight line.

4) Perceptron vs Delta / Gradient learning

The perceptron updates its parameters when the class is misclassified. But the delta rule updates parameters through the gradient, which is based on the magnitude of the prediction error.